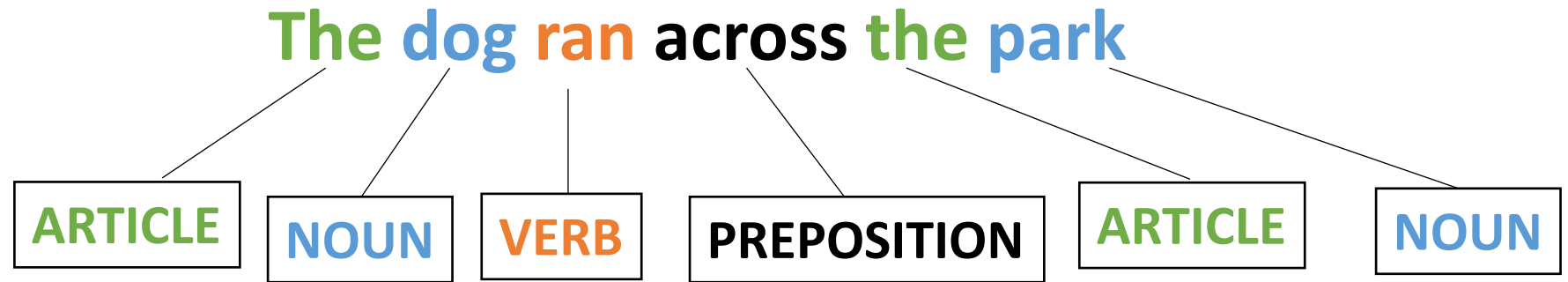


CSE110A: Compilers



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Naïve Scanner

simple string stream, peek/eat model

```
class NaiveScanner:
```

```
    def token(self):
```

```
        ...
```

```
        if self.ss.peek_char() in NUMS:
```

```
            value = ""
```

```
            while self.ss.peek_char() in NUMS:
```

```
                value += self.ss.peek_char()
```

```
                self.ss.eat_char()
```

```
            return ("NUM", value)
```

ID	=	[characters]
NUM	=	[numbers]
ASSIGN	=	"="
PLUS	=	"+"
MULT	=	"*"
IGNORE	=	[" "]

Shortcomings of Naïve scanner

- IDs with numbers in them?
 - `x1`, `y1`, `etc`.
 - how would you solve?
- Numbers with a decimal point in them?
 - `4.5`, `9999.99998`
 - how would you solve this?
- Two character operators:
 - `++`, `+=`
 - how would you solve this?

Review Topic:

Regular Expressions

Using REs

- any character `.`
- example using email (this is probably too general!)
 - `".*@\.*\.com"`
- What if we want either the domain or user name from the email?
- We can use groups!
 - use `()`s to delimitate groups
- `"(.*)@(\.*\.com)"`
- Index the resulting object with `[1]` and `[2]` to get to the user name and domain respectively

Using REs: Named Groups

- you can give **groups id names** rather than using indices
- “(?P<name>.+)@(?P<domain>.+ \.com)”

Let's write our tokens as regular expressions

- For our simple programming language

```
ID      = [characters]
NUM      = [numbers]
ASSIGN   = "="
PLUS     = "+"
MULT     = "*"
IGNORE   = [" ", "\n"]
```

```
ID      = "[a-z][a-z0-9]*"
NUM      = "[0-9]+"
ASSIGN   = "="
PLUS     = "+"
PP       = "++"
MULT     = "*"
IGNORE   = " | \n"
```

RE examples

- **What can REs not do?**
- Nested structures, such as parenthesis matching:
 - Try doing arithmetic expressions
 - You will not be able to match `()s`
- Classical example: REs cannot capture same number of repeats:
 - $A\{N\}B\{N\}$
- REs cannot parse HTML!!!
 - One of the most upvoted answers on stackoverflow!
 - <https://stackoverflow.com/questions/1732348/regex-match-open-tags-except-xhtml-self-contained-tags/1732454#1732454>

Review of RE Matchers to Build Scanner

- Exact match (EM) scanners
- Start-of-string (SOS) scanners
- named group (NG) scanners

EM Scanner

- Start with the whole string, remove one character at the end until a match is found. Then return the lexeme

ID	=	"[a-z]+"
NUM	=	"[0-9]+"
ASSIGN	=	"="
PLUS	=	"+"
MULT	=	"*"
IGNORE	=	" \n"
SEMI	=	";"

start with the whole string

"variable = 50 + 30 * 20;"

EM Scanner

- Start with the whole string, remove one character at the end until a match is found. Then return the lexeme

Try to match with all the tokens. No match.

ID	=	"[a-z]+"
NUM	=	"[0-9]+"
ASSIGN	=	"="
PLUS	=	"+"
MULT	=	"*"
IGNORE	=	" \n"
SEMI	=	";"

start with the whole string

"variable = 50 + 30 * 20;"

EM Scanner

- Start with the whole string, remove one character at the end until a match is found. Then return the lexeme

Try to match with all the tokens. No match.

ID	=	"[a-z]+"
NUM	=	"[0-9]+"
ASSIGN	=	"="
PLUS	=	"+"
MULT	=	"*"
IGNORE	=	" \n"
SEMI	=	";"

Try iteratively chopping characters from back

`"variable = 50 + 30 * 20;"`

EM Scanner

- Start with the whole string, remove one character at the end until a match is found. Then return the lexeme

Try to match with all the tokens. No match.

So on

ID	=	"[a-z]+"
NUM	=	"[0-9]+"
ASSIGN	=	"="
PLUS	=	"+"
MULT	=	"*"
IGNORE	=	" \n"
SEMI	=	";"

"variable = 50 + 30 * 20;"

EM Scanner

- Start with the whole string, remove one character at the end until a match is found. Then return the lexeme

Try to match with all the tokens. No match.

ID	=	"[a-z]+"
NUM	=	"[0-9]+"
ASSIGN	=	"="
PLUS	=	"+"
MULT	=	"*"
IGNORE	=	" \n"
SEMI	=	";"

Repeatedly delete characters from the end

"variable = 50 + 30 * 20;"

EM Scanner

- Start with the whole string, remove one character at the end until a match is found. Then return the lexeme

we can match id

ID	=	"[a-z]+"
NUM	=	"[0-9]+"
ASSIGN	=	"="
PLUS	=	"+"
MULT	=	"*"
IGNORE	=	" \n"
SEMI	=	";"

Until we find a match at this point

"variable = 50 + 30 * 20;"

Return the lexeme

(ID, "variable")

EM Scanner

- Pros
 - Uses an exact RE matcher. Many RE match algorithms are exact!
- Cons
 - SLOW! Each lexeme requires many many many calls to each RE match!

SOS (Start Of String) Scanner

- We will use a new RE match function

```
re.fullmatch(pattern, string, flags=0) ¶
```

If the whole *string* matches the regular expression *pattern*, return a corresponding [match object](#). Return `None` if the string does not match the pattern; note that this is different from a zero-length match.

```
re.match(pattern, string, flags=0)
```

If zero or more characters at the beginning of *string* match the regular expression *pattern*, return a corresponding [match object](#). Return `None` if the string does not match the pattern; note that this is different from a zero-length match.

SOS Scanner

- The match API gives us a match starting at the beginning of the string

Feed full string into each token definition

ID	=	"[a-z]+"
NUM	=	"[0-9]+"
ASSIGN	=	"="
PLUS	=	"+"
MULT	=	"*"
IGNORE	=	" \n"
SEMI	=	";"

"variable = 50 + 30 * 20;"

SOS Scanner

- The match API gives us a match starting at the beginning of the string

Feed full string into each token definition

ID	=	"[a-z]+"
NUM	=	"[0-9]+"
ASSIGN	=	"="
PLUS	=	"+"
MULT	=	"*"
IGNORE	=	" \n"
SEMI	=	";"

"variable = 50 + 30 * 20;"

We get 1 match. We can return the lexeme

(ID, "variable")

SOS Scanner

- The match API gives us a match starting at the beginning of the string

Chop the string

ID	=	"[a-z]+"
NUM	=	"[0-9]+"
ASSIGN	=	"="
PLUS	=	"+"
MULT	=	"*"
IGNORE	=	" \n"
SEMI	=	";"

"variable = 50 + 30 * 20;"

We get 1 match. We can return the lexeme

(ID, "variable")

SOS Scanner

- The match API gives us a match starting at the beginning of the string

Chop the string

ID	=	"[a-z]+"
NUM	=	"[0-9]+"
ASSIGN	=	"="
PLUS	=	"+"
MULT	=	"*"
IGNORE	=	" \n"
SEMI	=	";"

" = 50 + 30 * 20 ; "

We get 1 match. We can return the lexeme

(ID, "variable")

SOS Scanner

- The match API gives us a match starting at the beginning of the string

What about the next one

ID	=	"[a-z]+"
NUM	=	"[0-9]+"
ASSIGN	=	"="
PLUS	=	"+"
MULT	=	"*"
IGNORE	=	" \n"
SEMI	=	";"

" = 50 + 30 * 20 ; "

(ID, "variable")

SOS Scanner

- The match API gives us a match starting at the beginning of the string

What about the next one

ID	=	"[a-z]+"
NUM	=	"[0-9]+"
ASSIGN	=	"="
PLUS	=	"+"
MULT	=	"*"
IGNORE	=	" \n"
SEMI	=	";"

" = 50 + 30 * 20 ; "

1 match: IGNORE

(ID, "variable")

SOS Scanner

- The match API gives us a match starting at the beginning of the string

Chop the string

ID	=	"[a-z]+"
NUM	=	"[0-9]+"
ASSIGN	=	"="
PLUS	=	"+"
MULT	=	"*"
IGNORE	=	" \n"
SEMI	=	";"

" = 50 + 30 * 20 ; "

1 match: IGNORE

(ID, "variable")

SOS Scanner

- The match API gives us a match starting at the beginning of the string

Chop the string

ID	=	"[a-z]+"
NUM	=	"[0-9]+"
ASSIGN	=	"="
PLUS	=	"+"
MULT	=	"*"
IGNORE	=	" \n"
SEMI	=	";"

"= 50 + 30 * 20;"

1 match: IGNORE

(ID, "variable")

SOS Scanner

- Consideration

How to scan this string?

Try to match on each token

"CSE110A"

LETTERS	=	"[A-Z]+"
NUM	=	"[0-9]+"
CLASS	=	"CSE110A"

Two matches:

LETTERS: "CSE"

CLASS: "CSE110A"

Which one do we choose?

The longest one!

Why didn't we have to do this for the exact match Scanner?

*After each pass through token REs
we have to measure match length*

SOS Scanner

- One more consideration

Within 1 RE, how does this match?

```
CLASS = "CSE|110A|CSE110A"
```

"CSE110A"

SOS Scanner

- One more consideration

Within 1 RE, how does this match?

"CSE110A"

```
CLASS = "CSE|110A|CSE110A"
```

Returns "CSE", but this isn't what we want!!!

SOS Scanner

- One more consideration

Within 1 RE, how does this match?

"CSE110A"

```
CLASS = "CSE|110A|CSE110A"
```

Returns "CSE", but this isn't what we want!!!

When using the SOS Scanner: A token definition either should not:

- *contain choices where one choice is a prefix of another*
- *order choices such that the longest choice is the first one*

SOS Scanner

- One more consideration

Within 1 RE, how does this match?

"CSE110A"

```
CLASS = "CSE|110A|CSE110A"
```

Returns "CSE", but this isn't what we want!!!

When using the SOS Scanner: A token definition either should not:

- *contain choices where one choice is a prefix of another*
- *order choices such that the longest choice is the first one*

```
CLASS = "CSE110A|110A|CSE"
```

SOS Scanner

- Pros
 - Much faster than EM scanner. Only 1 call to each RE per `token()` call
- Cons
 - Depends on an efficient implementation of `match()`
 - Typically provided in most RE libraries (for this exact reason)
 - Requires some care in token definitions and prefixes

SOS Scanner

*We're going to optimize this to 1 RE call!
It can really help if you have many tokens*

- Pros
 - Much faster than EM scanner. Only 1 call to each RE per `token()` call
- Cons
 - Depends on an efficient implementation of `match()`
 - Typically provided in most RE libraries (for this exact reason)
 - Requires some care in token definitions and prefixes

NG (Named Group) Scanner

- We will still use the `match` API call

```
re.fullmatch(pattern, string, flags=0) ¶
```

If the whole *string* matches the regular expression *pattern*, return a corresponding [match object](#). Return `None` if the string does not match the pattern; note that this is different from a zero-length match.

```
re.match(pattern, string, flags=0)
```

If zero or more characters at the beginning of *string* match the regular expression *pattern*, return a corresponding [match object](#). Return `None` if the string does not match the pattern; note that this is different from a zero-length match.

NG Scanner

- Start out with token definitions
- Merge them into one RE definition

*Give each group a name
corresponding to its token*

ID	=	"[a-z]+"
NUM	=	"[0-9]+"
ASSIGN	=	"="
PLUS	=	"+"
MULT	=	"*"
IGNORE	=	" \n"
SEMI	=	";"

SINGLE_RE = "(?P<ID>[a-z]+) |
(?P<NUM>[0-9]+)
| (\. \.) |"

NG Scanner

- to implement `token()`

Try to match the whole string to the single RE

```
SINGLE_RE = "(?P<ID>[a-z]+) |  
            (?P<NUM>[0-9]+) |  
            (?P<ASSIGN>=) |  
            (?P<PLUS>+) |  
            (?P<MULT>*) |  
            (?P<IGNORE> |\\n) |  
            (?P<SEMI>;) "
```

```
"variable = 50 + 30 * 20;"
```

NG Scanner

- to implement `token()`

```
SINGLE_RE = "(?P<ID>[a-z]+) |  
            (?P<NUM>[0-9]+) |  
            (?P<ASSIGN>=) |  
            (?P<PLUS>+) |  
            (?P<MULT>*) |  
            (?P<IGNORE> |\\n) |  
            (?P<SEMI>;) "
```

Try to match the whole string to the single RE

```
"variable = 50 + 30 * 20;"
```

Check the `group` dictionary in the
result

NG Scanner

- to implement `token()`

```
SINGLE_RE = "(?P<ID>[a-z]+) |  
            (?P<NUM>[0-9]+) |  
            (?P<ASSIGN>=) |  
            (?P<PLUS>+) |  
            (?P<MULT>*) |  
            (?P<IGNORE> |\\n) |  
            (?P<SEMI>;) "
```

Try to match the whole string to the single RE

```
"variable = 50 + 30 * 20;"
```

```
{"ID"       : "variable"  
 "NUM"      : None  
 "ASSIGN"   : None  
 "PLUS"     : None  
 "MULT"     : None  
 "IGNORE"   : None  
 "SEMI"     : None}
```

NG Scanner

- to implement `token()`

Try to match the whole string to the single RE

```
SINGLE_RE = "(?P<ID>[a-z]+) |  
            (?P<NUM>[0-9]+) |  
            (?P<ASSIGN>=) |  
            (?P<PLUS>+) |  
            (?P<MULT>*) |  
            (?P<IGNORE> |\\n) |  
            (?P<SEMI>;) "
```

```
"variable = 50 + 30 * 20;"
```

```
{"ID"       : "variable"  
 "NUM"      : None  
 "ASSIGN"   : None  
 "PLUS"     : None  
 "MULT"     : None  
 "IGNORE"   : None  
 "SEMI"     : None}
```

Return the lexeme (ID, "variable")

NG Scanner

- to implement `token()`

chop!

```
SINGLE_RE = "(?P<ID>[a-z]+) |  
            (?P<NUM>[0-9]+) |  
            (?P<ASSIGN>=) |  
            (?P<PLUS>+) |  
            (?P<MULT>*) |  
            (?P<IGNORE> |\\n) |  
            (?P<SEMI>;) "
```

```
"variable = 50 + 30 * 20;"
```

```
{"ID"       : "variable"  
 "NUM"      : None  
 "ASSIGN"   : None  
 "PLUS"     : None  
 "MULT"     : None  
 "IGNORE"   : None  
 "SEMI"     : None}
```

Return the lexeme (ID, "variable")

NG Scanner

- to implement `token()`

chop!

```
SINGLE_RE = "(?P<ID>[a-z]+) |  
            (?P<NUM>[0-9]+) |  
            (?P<ASSIGN>=) |  
            (?P<PLUS>+) |  
            (?P<MULT>*) |  
            (?P<IGNORE> |\\n) |  
            (?P<SEMI>;) "
```

```
" = 50 + 30 * 20; "
```


How to deal with common prefixes in token definitions?

- Recall from SOS scanner:

LETTERS	=	"[A-Z]+"
NUM	=	"[0-9]+"
CLASS	=	"CSE110A"

How to scan this string?

"CSE110A"

How to deal with common prefixes in token definitions?

- Convert to a single RE

```
SINGLE_RE = "  
    (?P<LETTERS> ([A-Z] +) |  
    (?P<NUM> ([0-9] +) |  
    (?P<CLASS> CSE110A) "
```

How to scan this string?

"CSE110A"

How to deal with common prefixes in token definitions?

- Convert to a single RE

```
SINGLE_RE = "  
    (?P<LETTERS> ([A-Z] +) |  
    (?P<NUM> ([0-9] +) |  
    (?P<CLASS> CSE110A) "
```

How to scan this string?

"CSE110A"

What do we think the dictionary will look like?

How to deal with common prefixes in token definitions?

- Convert to a single RE

```
SINGLE_RE = "  
    (?P<LETTERS> ([A-Z] +) |  
    (?P<NUM> ([0-9] +) |  
    (?P<CLASS> CSE110A) "
```

How to scan this string?

"CSE110A"

```
{ "LETTERS" : "CSE"  
  "NUM"      : None  
  "CLASS"    : None  
}
```

How to deal with common prefixes in token definitions?

- Convert to a single RE

```
SINGLE_RE = "  
    (?P<LETTERS> ([A-Z] +) |  
    (?P<NUM> ([0-9] +) |  
    (?P<CLASS>CSE110A) "
```

"CSE110A"

```
{ "LETTERS" : "CSE"  
  "NUM"      : None  
  "CLASS"    : None  
}
```

What does this mean?

- Tokens should not contain prefixes of each other

OR

- Tokens that share a common prefix should be ordered such that the longer token comes first

How to deal with common prefixes in token definitions?

- Careful with these tokens

INCR	=	"++"
ADD	=	"+"
EQ	=	"=="
ASSIGN	=	"="

Ensure that you provide them in the right order so that the longer one is first!

NG Scanner

- Pros
 - FAST! Only 1 RE call per token ()
- Cons
 - Requires a named group RE library
 - inter-token interactions need to be considered

Token Actions: Functions as part of a token definition

- Once a token is matched, its token action is called on its lexeme,
- lexeme is then returned from scanner.

Token actions generally do three things:

- modify a value
 - e.g. change a type
- refine a token
 - e.g. ID -> reserved word
- modify the scanner state
 - e.g. return line and column number for error reporting.